

STATEMENT OF WORK

FOR

B-2 TEST PROGRAM SET

INTERFACE TEST ADAPTERS

**Software Directorate,
Tinker Air Force Base**

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1 SCOPE OF WORK

1.1 Scope of Work

The following establishes the minimum requirements for the design, testing, purchase, and delivery of Interface Test Adapters (ITAs) for use with the new TITAN receiver on the Common Organizational Level Tester (COLT). The Contractor shall develop and construct a first article to verify satisfactory ITA design and functionality. Full-rate production of the remaining ITAs, quantities described herein, is contingent upon successful Qualification Testing of the first articles by the Contractor and successful Acceptance Testing of the first articles by the Government Points of Contact (GPOCs). The minimum requirements for both types of testing and when they are to be executed are described in this document and its Appendices. When Acceptance Testing is successfully completed on any ITA, it shall be considered as delivered and become Government property. Upon successful acceptance of the final full-rate production ITA, the Contractor shall deliver to the Government all remaining technical data and physical items purchased for/relating to the design and production of the ITAs.

1.2 Background

For any and all uses of the phrase “Interface Test Adapter (ITA)” within this Statement of Work (SOW) and its Appendices; an “Interface Test Adapter (ITA)” shall be defined as the individual conglomerate electromechanical items that are named in Appendices A, B, and C, respectively. Within the B-2 Rotary Launcher Assembly (RLA) Test Program Set (TPS), B-2 Smart Bomb Rack Assembly (SBRA) TPS, B-2 Massive Ordnance Penetrator (MOP) Interface Test (MIT) TPS, and B-2 Bomb Rack Unit-44 (BRU-44) TPS there are older fielded versions of the ITAs. Each of the aforementioned TPSs use the COLT as their Automated Test System (ATS). The COLT, at the date of this document, uses proprietary MAC Panel MPX Connectors to interface with each TPS through an ITA. In the future, the COLT will be replacing the MPX Connectors with APEX Connectors and the TITAN Receiver—both of which are also MAC Panel proprietary products. In order to maintain continuity of service for the end user, the existing ITAs are to be replaced with the Contractor’s new ITAs and interface with the APEX Connectors and TITAN Receiver. The new ITAs shall be named, and from here on referred to as, the Powered RLA Interface Box (PRIB), SBRA Interface Box (SIB), and BRU-44 Cable (W1). The PRIB is to be used with the RLA TPS. The SIB is to be used with the SBRA TPS and the MIT TPS. The BRU-44 Cable is to be used with the BRU-44 TPS.

2 GOVERNMENT POINTS OF CONTACT

During the post-award phase, for any technical questions regarding the requirements in this SOW and its Appendices the Contractor shall contact the following Government Point of Contacts (GPOCs) within the Software Directorate. Communications to the GPOCs shall be by e-mail, but Technical Interchange Meetings can be hosted by the GPOCs over Microsoft Teams at the request of the Contractor or the Government. Any meetings hosted over Microsoft Teams by the GPOCs shall be recorded by the GPOCs. Should both of the GPOCs be unavailable for any period of time that exceeds five business days during the post-award phase, then they shall designate alternate technical points of contact within their unit and notify the Contractor by e-mail before such a time period would start. For any and all uses of the phrase “business days” within this Statement of Work and its Appendices; “business days” shall be defined as Monday through Friday 0800 – 1630 Central Standard Time, excluding federal holidays.

Primary Point of Contact:	Cade Guitron Electronics Engineer, NH-03 557 SWES/MXDPBA (405) 622-7174 cade.guitron@us.af.mil
Secondary Point of Contact:	Russell Prince Electronics Technician, NJ-03 557 SWES/MXDPBA (405) 622-7681 russell.prince.4@us.af.mil
Shipping Addressee:	Cade Guitron (405) 736-5350 4750 Staff Drive, Building 3220 Room 132, Tinker AFB, Oklahoma 73145

3 CONTRACTOR POINTS OF CONTACT

Within five business days of contract award, the Contractor shall provide to the GPOCs a Primary Point of Contact, Secondary Point of Contact, and shipping address for the production facility (CDRL 001, Appendix D). If the production facility is different than the testing facility for the ITAs, then the Contractor must also provide a shipping address, Primary Point of Contact, and Secondary Point of Contact for the testing facility. The Contractor must also provide the work e-mails for all of the aforementioned Contractor Point of Contacts (CPOCs). Should both of the CPOCs from the production facility be unavailable for any period of time that exceeds five business days during the post-award phase, then they shall designate alternate CPOCs within their facility and notify the GPOCs before such a time period would start.

4 APPLICABLE GOVERNMENT & INDUSTRY STANDARDS

The Contractor, at a minimum, is required to comply with the current editions of the following requirements for design, construction, and safety as applicable. The term “most recent edition” shall be understood to mean “most recently released edition up to the award date of this contract.”

4.1 Government Standards

- 4.1.1 MIL-DTL-5541F – Department of Defense Detail Specification, Chemical Conversion coatings on Aluminum and Aluminum Alloys, 1990 (R2006) or most recent edition.
- 4.1.2 MIL-PRF-28800G – Department of Defense Performance Specification, Test Equipment for Use with Electrical and Electronic Equipment, 1991 (R2021) or most recent edition.
- 4.1.3 MIL-PRF-32070A – Department of Defense Performance Specification, Test Program Sets, 2002 (R2012) or most recent edition.
- 4.1.4 MIL-PRF-8625F – Department of Defense Performance Specification, Anodic Coatings for Aluminum and Aluminum Alloy, 1968 (R2020) or most recent edition.

- 4.1.5 MIL-STD-130N – Department of Defense Standard Practice, Identification Marking of U.S. Military Property, 1993 (R2012) or most recent edition.
- 4.1.6 MIL-STD-1472H – Department of Defense Design Criteria Standard, Human Engineering, 1989 (R2020) or most recent edition.
- 4.1.7 MIL-STD-2035A (SH) – Department of Defense Test Method, Nondestructive Testing Acceptance Criteria, 1995 or most recent edition.
- 4.1.8 MIL-STD-810H – Department of Defense Test Method Standard, Environmental Engineering Considerations and Laboratory Tests, 1962 (R2022) or most recent edition.
- 4.1.9 (Application for copies should be addressed to Superintendent of Documents, Government Printing Office, Washington, D.C. 20402-001).

4.2 Industry Standards

- 4.2.1 American National Standard Institute (ANSI)
 - 4.2.1.1 ASTM SI 10 – American National Standard for use of the International System of Units (SI): The Modern Metric System, 2010 or most recent edition.
 - 4.2.1.2 (Application for copies should be addressed to American National Standards Institute, Attn: Sales Department, 1430 Broadway, New York, NY 10018-3363).
- 4.2.2 American Welding Society Incorporated (AWS)
 - 4.2.2.1 AWS D1.2D1.2M – Structural Welding Code—Aluminum, 2014 or most recent edition.
 - 4.2.2.2 (Applications for copies should be addressed to the American Welding Society Inc., 550 North LeJeune Road, Miami, FL 33126).
- 4.2.3 Global Electronics Association (IPC)
 - 4.2.3.1 IPC-A-610J – Acceptability of Electronic Assemblies, 2024 or most recent edition.
 - 4.2.3.2 IPC-J-STD-001F – Requirements for Soldered Electrical and Electronic Assemblies, 2016 or most recent edition.
 - 4.2.3.3 IPC/WHMA-A-620E – Requirements and Acceptance for Cable and Wire Harness Assemblies, 2022 or most recent edition.
 - 4.2.3.4 (Applications for copies should be addressed to the Global Electronics Association, 3000 Lakeside Drive, 105 North Bannockburn, IL 60015).
- 4.2.4 National Electrical Manufacturers Association (NEMA)
 - 4.2.4.1 NEMA ICS 6 – Industrial Control and Systems: Enclosures, 2006 or most recent edition.
 - 4.2.4.2 (Application for copies should be addressed to National Electrical Manufacturers Association, 1300 N 17th Street, Suite 1752, Rosslyn, VA 22209).

4.3 Order of Precedence

In the event of a conflict between any requirement in this Statement of Work and the references cited herein, the requirement within this Statement of Work takes precedence. However, nothing in this document or its Appendices supersedes applicable laws and regulations unless a specific exception has been obtained.

5 ITA DESIGN REQUIREMENTS

5.1 Government Document Package

The following drawings shall be provided to the Contractor to form the basis for their ITA designs. Should the Contractor identify any potential issues that would impact the production of the ITAs in the government provided documents, then they shall present them to the GPOCs no later than Design Review.

- 5.1.1 BRU-44 Cable Electrical Drawing & Parts List (Appendix A)
- 5.1.2 PRIB Electrical Drawing & Parts List (Appendix B)
- 5.1.3 SIB Electrical Drawing & Parts List (Appendix C)

5.2 General Requirements

- 5.2.1 COLT Interface – The COLT is a box-shaped ATS, with dimensions 22”x22”x22”, that will interface with TPSs through a MAC Panel TITAN Receiver.
 - 5.2.1.1 The BRU-44 Cable, PRIB, and SIB must all be able to interface with the TITAN Receiver without impacting access to any components on the COLT Top Face—with a minimum of 4” clearance given between the COLT Top Face and any part of the ITA that does not directly interface with the TITAN Receiver.
- 5.2.2 Dimensional Systems – U.S. Customary System of Units (US) shall be used in the design, documentation, and construction of all ITAs. In this SOW and its Appendices, all measurements, dimensions, sizes, and capacities are given in the U.S. Customary System of Units (US).
- 5.2.3 Documentation Markings – All documents, drawings, and schematics that are delivered by the contractor must have every page marked with CUI on the top of the page and on the bottom of the page. Each document, drawing, and schematic must also be designated as Distribution Statement D and have it marked on each of their respective first/cover pages.
- 5.2.4 Electrical Diagrams – The electrical diagrams within Appendices A, B, and C must be matched exactly from pin to pin when designing and constructing the ITAs. Any deviation from such may result in damage to existing equipment. The goal of this requirement is for the Contractor’s ITAs to replace the existing ITAs without any interruption in service for the end user and for no modifications to be needed for the ITAs’ respective TPSs.
- 5.2.5 Environmentally Sealed – All ITAs will be used in all types of environments and as such, must be hermetically sealed to prevent dust, water, sand, or other Foreign Object Debris (FOD) from entering the ITA during transit or while in operational use.
 - 5.2.5.1 Within this SOW, FOD shall be defined as any and all physical or chemical items that may inappropriately exist on or within the ITA that can cause damage to equipment or injure personnel—i.e. anything including, but not limited to, flux residue, metal shavings, loose screws, wire scraps, dirt, et cetera.

- 5.2.6 Parts List – Within each drawing, there are parts for which there exists no suitable substitutes. They must be used in the design of the new ITAs and shall be marked on the drawing with a “Find Note.” For all other parts, suitable substitutes may exist, but the Contractor must acquire approval from the GPOCs at Design Review before using them in the construction of ITAs.
- 5.2.7 Parts Procurement – All parts purchased by the Contractor for the purpose of designing and constructing the ITAs must be sourced domestically or from Trade Agreement Act compliant countries.
- 5.2.8 Soldering – All soldering performed for the purpose of constructing the ITAs must use SN63PB37-type or SN60PB40-type solder.

5.3 Specific Requirements: BRU-44 Cable

The BRU-44 Cable requires no redesign. It shall be built as is specified in Appendix A.

5.4 Specific Requirements: PRIB

Appendix B provides the electrical drawings and parts lists to be used in the Contractor’s design of the PRIB. Any notes written in Appendix B are requirements for the design of the PRIB. All other specific requirements for design are listed in this section.

- 5.4.1 Data Plate – A data plate shall be affixed to an external face of the SIB.
 - 5.4.1.1 The text on the data plate shall be at least 1/8 of an inch tall and no larger than 1/4 of an inch tall.
 - 5.4.1.2 It must be placed in such a way that its view is not obstructed during the operational use of the ITA.
 - 5.4.1.3 It must not take up more than 16 square-inches on the face it is affixed to.
 - 5.4.1.4 It shall contain information about the ITA to include:
 - 5.4.1.4.1 Name of the ITA
 - 5.4.1.4.2 Contractor’s Part Number for the ITA
 - 5.4.1.4.3 Serial Number of the ITA
 - 5.4.1.4.4 Revision Letter of the ITA
- 5.4.2 Dimensions – The PRIB, and all of its constituent parts, must not spatially exceed a length by width of 11”x11” nor shall it exceed 10” of height.
 - 5.4.2.1 Additional Considerations – No part of the PRIB shall spatially exceed the lever action arm of the TITAN Receiver nor shall it exceed the area of the top face of the COLT. Details of the configuration of the COLT top face shall be provided by the GPOCs no later than five business days after the CPOCs provide their contact information to the GPOCs.
- 5.4.3 Enclosure – The walls of the enclosure containing the constituent parts of the PRIB shall be made of one of two materials only:
 - 5.4.3.1 5052 – H32 Aluminum Alloy may be used only for walls that do not have connectors affixed to them.

- 5.4.3.2 6061 – T6 Aluminum Alloy must be used for any and all walls of the enclosure that have connectors affixed to them.

5.4.4 External Components Placement

- 5.4.4.1 Connectors J5 – J12 must all be grouped together and affixed to the same face of the enclosure.
- 5.4.4.2 Connectors J13 – J19 must all be grouped together and affixed to the same face of the enclosure.
- 5.4.4.3 Connectors J5 – J19 shall not be placed closer together than 3.5” when measured from center to center of each connector.
- 5.4.4.4 All circuit breakers (CB) and LEDs (D) must be placed on the same face of the enclosure and externally accessible/visible.
- 5.4.4.5 Each set of three LEDs must be grouped immediately next to its respective breaker as follows:
 - 5.4.4.5.1 CB1 & D1 – D3
 - 5.4.4.5.2 CB2 & D4 – D6
 - 5.4.4.5.3 CB3 & D7 – D9
 - 5.4.4.5.4 CB4 & D10 – D12
 - 5.4.4.5.5 CB5 & D13 – D15
 - 5.4.4.5.6 CB6 & D16 – D18
 - 5.4.4.5.7 CB7 & D19 – D21
 - 5.4.4.5.8 CB8 & D22 – D24
 - 5.4.4.5.9 CB9 & D25 – D27

5.4.5 Finish – Only two types of finish may be used on the walls of the enclosure:

- 5.4.5.1 If a wall is made of 5052 – H32 Aluminum Alloy, then it must be anodized per MIL-PRF-8625F, Type II, Class I and then the outward facing side of the wall must be powder coated.
- 5.4.5.2 If a wall is made of 6061 – T6 Aluminum Alloy, then it must be finished with gold Alodine per MIL-DTL-5541F, Class 1A and then the outward facing side of the wall must be powder coated.

5.4.6 Labeling – All components that are affixed to the enclosure and externally visible shall be labeled through etching and filling with an ink that is high contrast when compared to the color of the enclosure.

- 5.4.6.1 Every connector shall be labeled with its connector designation as written in Appendix B—e.g. J1, J2, J101, et cetera.
- 5.4.6.2 Every set of circuit breakers and LEDs must be given one label per grouping as follows:
 - 5.4.6.2.1 CB1/D1 – D3: “NWCM 2”
 - 5.4.6.2.2 CB2/D4 – D6: “NWCM 1”
 - 5.4.6.2.3 CB3/D7 – D9: “SQUIB BU”
 - 5.4.6.2.4 CB4/D10 – D12: “SQUIB PRI”

- 5.4.6.2.5 CB5/D13 – D15: “MSRA (ODD)”
- 5.4.6.2.6 CB6/D16 – D18: “MSRA (EVEN)”
- 5.4.6.2.7 CB7/D19 – D21: “TRU ELECT 2”
- 5.4.6.2.8 CB8/D22 – D24: “TRU HYD”
- 5.4.6.2.9 CB9/D25 – D27: “TRU ELECT 1”

- 5.4.7 Ruggedization – The end user for the PRIB are airmen in both environmentally controlled buildings and outdoors in a wide variety of environments. The Contractor must take this into consideration and determine the best configuration of constituent parts and additions to the design that prevent catastrophic damage to the enclosure when dropped—e.g. a crack or dent forming in the enclosure that would impact the functionality of the PRIB during operational use or prevent repairs.
- 5.4.8 Shape – The overall shape of the SIB and its constituent parts shall be that of a rectangular box, rectangular prism, or cuboid.
- 5.4.9 Weight – The total weight of the PRIB shall not exceed 100 lbs.

5.5 Specific Requirements: SIB

Appendix C provides the electrical drawings and parts lists to be used in the Contractor’s design of the SIB. Any notes listed in Appendix C are requirements for the design of the SIB. All other specific requirements for design are listed in this section.

- 5.5.1 Data Plate – A data plate shall be affixed to an external face of the SIB.
 - 5.5.1.1 The text on the data plate shall be at least 1/8 of an inch tall and no larger than 1/4 of an inch tall.
 - 5.5.1.2 It must be placed in such a way that its view is not obstructed during the operational use of the ITA.
 - 5.5.1.3 It must not take up more than 16 square-inches on the face it is affixed to.
 - 5.5.1.4 It shall contain information about the ITA to include:
 - 5.5.1.4.1 Name of the ITA
 - 5.5.1.4.2 Contractor’s Part Number for the ITA
 - 5.5.1.4.3 Serial Number of the ITA
 - 5.5.1.4.4 Revision Letter of the ITA
- 5.5.2 Dimensions – The SIB, and all of its constituent parts, must not spatially exceed a length by width of 11”x11” nor shall it exceed 10” of height.
 - 5.5.2.1 Additional Considerations – No part of the SIB shall spatially exceed the lever action arm of the TITAN Receiver nor shall it exceed the area of the top face of the COLT.
- 5.5.3 Enclosure – The walls of the enclosure containing the constituent parts of the PRIB shall be made of one of two materials only:
 - 5.5.3.1 5052 – H32 Aluminum Alloy may be used only for walls that do not have connectors affixed to them.

- 5.5.3.2 6061 – T6 Aluminum Alloy must be used for any and all walls of the enclosure that have connectors affixed to them.
- 5.5.4 External Components Placement
 - 5.5.4.1 Connectors J5 – J18 shall not be placed closer together than 3.5” when measured from center to center of each connector.
- 5.5.5 Finish – Only two types of finish may be used on the walls of the enclosure:
 - 5.5.5.1 If a wall is made of 5052 – H32 Aluminum Alloy, then it must be anodized per MIL-PRF-8625F, Type II, Class I and then the outward facing side of the wall must be powder coated.
 - 5.5.5.2 If a wall is made of 6061 – T6 Aluminum Alloy, then it must be finished with gold Alodine per MIL-DTL-5541F, Class 1A and then the outward facing side of the wall must be powder coated.
- 5.5.6 Labeling – All components that are affixed to the enclosure and externally visible shall be labeled through etching and filling with an ink that is high contrast when compared to the color of the enclosure.
 - 5.5.6.1 Every connector shall be labeled with its connector designation as written in Appendix C—e.g. J1, J2, J3, et cetera.
- 5.5.7 Ruggedization – The end user for the SIB are airmen in both environmentally controlled buildings and outdoors in a wide variety of environments. The Contractor must take this into consideration and determine the best configuration of constituent parts and additions to the design that prevent catastrophic damage to the enclosure when dropped—e.g. a crack or dent forming in the enclosure that would impact the functionality of the SIB during operational use or prevent repairs.
- 5.5.8 Shape – The overall shape of the SIB and its constituent parts shall be that of a rectangular box, rectangular prism, or cuboid.
- 5.5.9 Weight – The total weight of the SIB shall not exceed 100 lbs.

6 DESIGN REVIEW

No more than forty-five business days after contract award, an online Microsoft Teams meeting shall be hosted by the GPOCs. During which, the Contractor shall give a presentation (CDRL 003, Appendix F) on their proposed designs. In addition to the presentation, the Contractor shall provide the GPOCs a Preliminary Technical Data Package (CDRL 002, Appendix E). The Contractor must send the Preliminary Technical Data Package to the GPOCs through DoD-Safe no less than five business days before the Design Review.

6.1 Preliminary Technical Data Package

The Preliminary Technical Data Package must include the following:

- 6.1.1 BRU-44 Cable Drawings
 - 6.1.1.1 Mechanical drawings of the BRU-44 Cable in *.dwg* or *.dxf* file formats, which include the following:

- 6.1.1.1.1 Dimensional measurements of the BRU-44 Cable and its constituent parts on every drawing.
- 6.1.1.1.2 To-scale 2D line drawing of a side view of the BRU-44 Cable (see Appendix A for reference drawing).
- 6.1.1.2 Electrical drawings of the BRU-44 Cable in *.dwg* or *.dxf* file formats, which include the following:
 - 6.1.1.2.1 Pin-to-pin wire paths with full labeling/naming of every pin that is used.
 - 6.1.1.2.2 Correct referencing of part and drawing numbers.
- 6.1.2 PRIB Drawings
 - 6.1.2.1 Mechanical drawings of the PRIB in *.dwg* or *.dxf* file formats, which include the following:
 - 6.1.2.1.1 Dimensional measurements of the PRIB and its constituent parts on every drawing.
 - 6.1.2.1.2 To-scale 3D model of the external faces of the entire PRIB populated with its subcomponents.
 - 6.1.2.1.3 To-scale 3D model of the external faces of the entire PRIB not populated with its subcomponents.
 - 6.1.2.1.4 To-scale individual 2D cutouts of each face of the PRIB.
 - 6.1.2.1.5 A parts list for every individual item needed to construct the PRIB in its entirety—to include, but not limited to, consumable items such as solder, zip ties, screws, et cetera.
 - 6.1.2.1.6 Any notes relevant to the construction process of the PRIB that would be necessary to reproduce the PRIB exactly as the Contractor designed.
 - 6.1.2.2 Electrical drawings of the PRIB in *.dwg* or *.dxf* file formats, which include the following:
 - 6.1.2.2.1 Pin-to-pin wire paths with full labeling/naming of every pin that is used.
 - 6.1.2.2.2 Correct referencing of part and drawing numbers.
 - 6.1.2.3 Gerber files (*.gbr* file format) for any and all non-commercial off the shelf (non-COTS) Printed Circuit Boards (PCBs) that are designed for and used in the PRIB.
- 6.1.3 SIB Drawings
 - 6.1.3.1 Mechanical drawings of the SIB in *.dwg* or *.dxf* file formats, which include the following:
 - 6.1.3.1.1 Dimensional measurements of the SIB and its constituent parts on every drawing.
 - 6.1.3.1.2 To-scale 3D model of the external faces of the entire SIB populated with its subcomponents.
 - 6.1.3.1.3 To-scale 3D model of the external faces of the entire SIB not populated with its subcomponents.
 - 6.1.3.1.4 To-scale individual 2D cutouts of each face of the SIB.
 - 6.1.3.1.5 A parts list for every individual item needed to construct the SIB in its entirety—to include, but not limited to, consumable items such as solder, zip ties, screws, et cetera.
 - 6.1.3.1.6 Any notes relevant to the construction process of the SIB that would be necessary to reproduce the SIB exactly as the Contractor designed.
 - 6.1.3.2 Electrical drawings of the SIB in *.dwg* or *.dxf* file formats, which include the following:
 - 6.1.3.2.1 Pin-to-pin wire paths with full labeling/naming of every pin that is used.

6.1.3.2.2 Correct referencing of part and drawing numbers.

6.1.3.3 Gerber files (.gbr file format) for any and all non-commercial off the shelf (non-COTS) Printed Circuit Boards (PCBs) that are designed for and used in the SIB.

6.2 Design Review Presentation

The presentation shall be in PowerPoint (.pptx) Format, be e-mailed to the GPOCs on the same business day as Design Review, and, at the least, include the following:

- 6.2.1 An overall project schedule that, at the least, outlines lead time for parts procurement, production schedule for the first articles of each ITA, and a production schedule for each full-rate production ITA.
- 6.2.2 Any concerns identified with the government provided documents (Appendices A, B, and C) that would impact the design and construction of the ITAs; such as unclear instructions, long parts acquisition lead-times, or unavailable parts.
- 6.2.3 A test plan for the Temperature Testing Event—described in SOW Section 9—that includes the following:
 - 6.2.3.1 Location – The facility name and address where the testing event will occur.
 - 6.2.3.2 Proposed Dates – Testing must occur during consecutive weekdays and at least two sets of dates, that occur on separate weeks and do not include a federal holiday, must be proposed by the Contractor as options for the Government to choose from. If the test event is scheduled to last more than five weekdays, then the testing must also occur during consecutive weeks.
 - 6.2.3.3 Proposed Schedule – The schedule must include the hours of work during each day, tests expected to be accomplished within each day, and the total duration of the event.
 - 6.2.3.4 Other Details – How sensor data will be collected and reported and where the sensors will be placed within the chamber.
- 6.2.4 Any and all information that the Contractor believes is pertinent for the GPOCs to know or consider about the design, construction, and testing of the ITAs.

6.3 Post Design Review

The GPOCs shall have no more than ten business days from the Design Review date to review and provide an e-mail response (approval or disapproval) to the Preliminary Technical Data Package and items within the Design Review Presentation. As well as to address any inquiries the Contractor may have raised about parts procurement or other subjects.

- 6.3.1 GPOC Approval – If the GPOCs identify no concerns or adjustments to the proposed designs of the ITAs, then they shall sign off on the Preliminary Technical Data Package and Project Schedule.

- 6.3.2 GPOC Disapproval – If the GPOCs find anything that requires adjustment in the Contractor’s Preliminary Data Package, they will detail them in their response e-mail. Any notes, corrections, or changes to the Contractor’s drawings shall be annotated onto the drawings directly and attached to the e-mail. The Contractor shall then have ten business days to respond by e-mail. At any point during this process the Contractor or GPOCs may request a Technical Interchange Meeting to discuss talking points in the e-mails. This process shall continue until GPOC Approval is reached.

7 PRODUCTION

Should the contractor identify potential issues with the production schedule described in this section—due to parts lead-times, supply chain problems, manufacturing capabilities, or any similar reasons—then they shall present them to the GPOCs no later than Design Review. Should unforeseen issues arise with any of the aforementioned after Design Review, then the Contractor must notify the GPOCs through e-mail no later than five business days after the identification of the problem. This will allow for adjustment of the production and testing schedules for both the GPOCs and the Contractor.

7.1 First Articles

- 7.1.1 BRU-44 Cable – The Contractor shall produce a first article of the BRU-44 Cable within sixty business days of the GPOCs’ acceptance of the Contractor’s proposal post Design Review
- 7.1.2 PRIB – The Contractor shall produce a first article of the PRIB within sixty business days of the GPOCs’ acceptance of the Contractor’s proposal post Design Review.
- 7.1.3 SIB – The Contractor shall produce a first article of the SIB within sixty business days of the GPOCs’ acceptance of the Contractor’s proposal post Design Review.

7.2 Full-Rate Production

- 7.2.1 BRU-44 Cable – Starting the month following successful acceptance of the first article of the BRU-44 Cable, the Contractor shall produce one BRU-44 Cable every month—before the end of the final business day of the month—until eight total full-rate production BRU-44 Cable are produced.
- 7.2.2 PRIB – Starting the month following successful acceptance of the first article of the PRIB, the Contractor shall produce one PRIB every month—before the end of the final business day of the month—until nine total full-rate production PRIBs are produced.
- 7.2.3 SIB – Starting the month following successful acceptance of the first article of the SIB, the Contractor shall produce one SIB every month—before the end of the final business day of the month—until thirteen total full-rate production SIBs are produced.

8 QUALIFICATION TESTING

The Contractor, at a minimum, must meet the following requirements when testing the completely constructed ITAs. The Contractor shall complete Qualification Testing of an ITA within ten business days of the ITA's production date with exception to the first articles of the PRIB and SIB, whose Qualification Testing is dependent upon the successful completion of the Temperature Testing Event.

8.1 Electrical Testing

The Contractor shall complete the following electrical tests on all ITAs.

- 8.1.1 Continuity Testing – The Contractor shall verify with a Digital Multimeter (DMM) that continuity of less than or equal to 5 Ohms exists from end to end of every individual wire path in accordance with (IAW) the ITA's respective schematic—unless otherwise specified by the ITA's respective schematic.
 - 8.1.1.1 Where otherwise specified, the Contractor shall verify with a Digital Multimeter (DMM) that the Ohm value of the end-to-end wire path matches the Ohm value specified on the schematic with a margin of error of plus or minus 5%.
- 8.1.2 Isolation Testing – The Contractor shall verify with a DMM that no continuity exists between any connector's pins and its own shell—unless otherwise specified by the ITA's respective schematic.
 - 8.1.2.1 Where otherwise specified, the Contractor shall verify with a Digital Multimeter (DMM) that the Ohm value of the end-to-end wire path matches the Ohm value specified on the schematic with a margin of error of plus or minus 5%.

8.2 Environmental Testing

The Contractor shall complete the following environmental tests on only the first articles of the PRIB and SIB. Government Furnished Equipment (GFE) is required for the temperature testing and the GPOC's must be onsite to operate the GFE. SOW Section 9 details the required GFE, required personnel, and definitions of successful/unsuccessful temperature testing.

- 8.2.1 High Temperature Testing – The first article of the PRIB must receive the following Temperature Tests outlined MIL-STD-810H w/CHANGE 1 METHOD 501.7:
 - 8.2.1.1 Operation (Procedure II) – Section 4.5.3 (With the following input parameters and the PRIB installed onto the COLT.)
 - 8.2.1.1.1 Exposure Type I – Constant 23C for two hours, with the COLT operating and running the Mini-Mockup Simulator (MMS) TPS.
 - 8.2.1.1.2 Exposure Type II – Constant 23C for four hours, with the COLT not operating and powered off.
 - 8.2.1.1.3 Exposure Type III – Constant 49C for four hours, with relative humidity below 20%, the COLT operating, and the COLT running the MMS TPS.
 - 8.2.1.1.4 Inspection Criteria – For each exposure type, SOW Sections 8.3.2, 8.3.3, 8.4.2, 8.4.3, 8.4.5, 8.4.6.
 - 8.2.1.1.5 Operational Check Criteria – For each exposure type, MMS TPS and SOW Section 8.1.

- 8.2.1.1.6 Note: While the first article of the SIB shall not undergo temperature testing—it cannot move into Acceptance Testing until the PRIB passes temperature testing.
- 8.2.2 Low Temperature Testing – The first article of the PRIB must receive the following Temperature Tests outlined MIL-STD-810H w/CHANGE 1 METHOD 502.7:
 - 8.2.2.1 Operation (Procedure II) – Section 4.5.3 (With the following input parameters and the PRIB installed onto the COLT.)
 - 8.2.2.1.1 Exposure Type I – Constant -40C for four hours, with the COLT not operating and powered off.
 - 8.2.2.1.2 Exposure Type II – Constant -51C for four hours, with the COLT not operating and powered off.
 - 8.2.2.1.3 Inspection Criteria – For each exposure type, SOW Sections 8.3.2, 8.3.3, 8.4.2, 8.4.3, 8.4.5, 8.4.6.
 - 8.2.2.1.4 Operational Check Criteria – For each exposure type, MMS TPS and SOW Section 8.1.
 - 8.2.2.1.5 Note: While the first article of the SIB shall not undergo temperature testing—it cannot move into Acceptance Testing until the PRIB passes temperature testing.
- 8.2.3 Shock Testing – The PRIB and SIB shall receive the following Shock Tests outlined in MIL-STD-810H w/CHANGE 1 METHOD 516.8:
 - 8.2.3.1 Bench Handling (Procedure VI) – Section 4.6.7 (With the following input parameters.)
 - 8.2.3.1.1 Inspection Criteria – SOW Sections 8.3.1, 8.3.2, 8.4.4, 8.4.5, 8.4.6.
 - 8.2.3.1.2 Operational Check Criteria – None.
 - 8.2.3.2 Transportation Shock (Procedure II) – Section 4.6.3 (With the following input parameters.)
 - 8.2.3.2.1 Table 516.8-VII. On Road and Off Road.
 - 8.2.3.2.2 Inspection Criteria – SOW Sections 8.3.1, 8.3.2, 8.4.4, 8.4.5, 8.4.6.
 - 8.2.3.2.3 Operational Check Criteria – SOW Qualification Testing Section 8.1.
 - 8.2.3.2.4 Unpackaged Handling – ITA must be tested outside of its transit case.
 - 8.2.3.3 Transit Drop (Procedure IV) – Section 4.6.5 (With the following input parameters.)
 - 8.2.3.3.1 Table 516.8-X. Tactical Transport Drop Test.
 - 8.2.3.3.2 Inspection Criteria – SOW Sections 8.3.1, 8.3.2, 8.4.4, 8.4.5, 8.4.6.
 - 8.2.3.3.3 Operational Check Criteria – Damage of any kind to the enclosure that would prevent the mating of cables to the ITA, prevent the mating of the ITA to the COLT, or result in the potential for FOD accessing the inside of the enclosure, shall constitute the failure criteria for an operational check. Damage or non-damage of any other kind shall constitute the pass criteria for an operational check.
 - 8.2.3.3.4 Unpackaged Handling – ITA must be tested outside of its transit case.
 - 8.2.3.3.5 Bottom – Face with APEX connectors (Do NOT drop on this face/execute this parameter).
 - 8.2.3.3.6 Two Faces – Any face without MIL-SPEC connectors or breakers/lights.
 - 8.2.3.3.7 Two Edges – Any edges.

8.3 Physical Testing

The Contractor shall complete the following physical tests on all ITAs.

- 8.3.1 The Contractor shall verify that all pins are all fully inserted into the locking mechanism of all MIL-STD Connectors and cannot recede when pulled on from the back by hand.
- 8.3.2 The Contractor shall verify that all screws and terminal connections are all fully seated and secured.
- 8.3.3 The Contractor shall verify that all labels and etchings do not incur damage when touched by a person.

8.4 Visual Inspections

The Contractor shall complete the following visual inspections on all ITAs.

- 8.4.1 The Contractor shall verify that all solder joints meet the acceptability requirements detailed in IPC-A-610J Section 5: Soldering.
- 8.4.2 The Contractor shall verify that all wires meet the acceptability requirements detailed in IPC-A-610J Section 6: Terminal Connections.
- 8.4.3 The Contractor shall verify that all components within the ITA meet the acceptability requirements detailed in IPC-A-610J Section 10.6: Cleanliness.
- 8.4.4 The Contractor shall verify that all pins in all connectors are not bent, missing, or otherwise damaged.
- 8.4.5 The Contractor shall verify that all labels and text seen on or in the ITA are legible and undamaged.
- 8.4.6 The Contractor shall verify that no FOD exists within the ITA.

8.5 Qualification Test Reports (CDRL 004, Appendix G)

8.5.1 General Requirements

- 8.5.1.1 The Contractor shall generate Qualification Test Reports for all ITAs.
- 8.5.1.2 The test report must explicitly state what tests were performed on the ITA.
- 8.5.1.3 Each test shall be given an explicit title and clearly labeled with the result of the test as “PASSED” or “FAILED,” with an overall “PASSED” result only given to the ITA when all of the listed tests pass.
- 8.5.1.4 When a test fails, details of the ITA’s failure shall be written under the appropriate test header and the details of the repair for the ITA shall be written under the same test header.
- 8.5.1.5 Multiple Qualification Test Reports shall not be generated for an individual ITA. Anytime an ITA fails a test and is repaired the new tests that are performed shall be appended to the end of the ITA’s Qualification Test Report, thus creating a continuous history of the ITA as it is tested by the Contractor.

- 8.5.1.6 An ITA shall only be shipped to the GPOCs if and only if the overall result of its respective Qualification Test Report is “PASSED.”
- 8.5.1.7 A digital copy of the Test Report shall be sent through e-mail to the GPOCs within one business day of the ITA successfully completing testing.
- 8.5.2 Qualification Testing Pass Criteria – When an ITA passes every test during the Contractor’s Qualification Testing, the ITA shall be considered ready for Acceptance Testing and shipped to the GPOCs within ten business days.
 - 8.5.2.1 Whenever the Contractor ships ITAs to the GPOCs, they must also provide a tracking number from the shipping company through e-mail no later than one business day after ITA shipment.
- 8.5.3 Qualification Testing Failure Criteria – For any test failures that occur during the Temperature Testing Event refer to SOW Section 9. When an ITA fails any other test during Qualification Testing, the Contractor shall then perform the appropriate repairs within ten business days—from the date of failure—on the ITA to fix what caused the failure. They shall then retest the ITA within ten business days of the repair and repeat this process until the ITA completely passes Qualification Testing.

9 TEMPERATURE TESTING EVENT

In order to accomplish the temperature testing portions of the Environmental Testing detailed in SOW Sections 8.2.1 and 8.2.2, the Contractor must have GFE to accomplish the testing and the GPOCs present to operate the GFE. A Temperature Testing Event shall be hosted by the Contractor to facilitate the completion of Environmental Testing.

9.1 Required GFE

The following GFE will be, at the minimum, required to be shipped to the Contractor’s Testing Facility within five business days before the week of temperature testing.

- 9.1.1 COLT – The COLT is needed to run the MMS TPS.
- 9.1.2 COLT Accessory Kit – The Accessory Kit contains the PDCU and self-test adapter for the COLT.
- 9.1.3 MMS TPS – The MMS TPS is for-engineering-use-only software that is based on the RLA TPS. It requires only the PRIB, PDCU, and COLT to operate. Further details of the MMS TPS shall be provided to the CPOCs by the GPOCs no later than five business days after the CPOCs are known to the GPOCs.
- 9.1.4 Portable Display Control Unit (PDCU) – The PDCU is required to interface with and operate the COLT.

9.2 Required Personnel

Government and contractor personnel must both be present at the Temperature Testing Event.

- 9.2.1  At a minimum, the following government personnel—or their selected alternates—must be present during temperature testing:
 - 9.2.1.1 GPOCs

- 9.2.2 At the minimum, the following contractor personnel—or their selected alternates—must be present during temperature testing:
 - 9.2.2.1 Primary CPOC for the manufacturing facility.
 - 9.2.2.2 Primary CPOC for the testing facility, if different from the manufacturing facility.
 - 9.2.2.3 Contractor Technical Expert that is trained to operate the temperature chamber and execute the test plan.

9.3 Test Event Results

- 9.3.1 Non-ITA Failures – Should the COLT or PDCU fail during any part of the Temperature Testing Event, it shall not constitute an operational failure/fault on the part of the PRIB. Any failures that may be the result of the COLT or PDCU shall be identified and, if possible, fixed onsite by the CPOCs so that testing may resume. Otherwise, the test event shall be rescheduled or delayed for a functioning COLT or PDCU to be shipped onsite by the Government.
- 9.3.2 Successful Event Criteria – The PRIB passes all temperature tests without any failures. At which point, both the first articles of the PRIB and SIB shall be designated as having passed Qualification Testing.
- 9.3.3 Unsuccessful Event Criteria – The PRIB fails any of the Inspection or Operational Check Criteria defined in SOW Sections 8.2.1 and 8.2.2.
 - 9.3.3.1 The Contractor shall have twenty business days to develop a solution to the failure and implement it onto the PRIB. During this time the Contractor and the Government shall work together to schedule a new test event that shall occur no later than a month after the date the solution is implemented. This process shall be repeated until the test event is successful.
 - 9.3.3.2 Any modifications that are made to the PRIB to pass the Temperature Testing Event must also be performed on the first article of the SIB and then every Qualification Test (except for Environmental Testing) must be re-executed on the SIB. Once both the newly modified first articles of the PRIB and SIB pass Acceptance Testing, their designs shall become the ones used for Full-Rate Production.

10 **ACCEPTANCE TESTING**

The GPOCs shall perform similar testing to the Contractor within ten business days of its receipt. The exception being that the GPOCs shall execute TPS Testing on the ITAs in lieu of Environmental Testing. In the event of quality control issues, the GPOCs reserve the right to change any tests that are listed as being performed on “only the first articles of the ITAs” to being performed on “all ITAs.”

10.1 Electrical Testing

The GPOCs shall complete the same electrical tests as the Contractor on only the first articles of the ITAs.

10.2 Physical Testing

The GPOCs shall complete the same physical tests as the Contractor on only the first articles of the ITAs.

10.3 TPS Testing

The GPOCs shall run the following TPSs with all ITAs. Details of the tests involved with each TPS shall be provided by the GPOCs to the CPOCs after contract award; no later than five business days after the CPOCs provide their contact information to the GPOCs.

- 10.3.1 The GPOCs shall run an End-to-End for every available test in the BRU-44 TPS with the BRU-44 Cable.
- 10.3.2 The GPOCs shall run an End-to-End for every available test in the RLA TPS with the PRIB.
- 10.3.3 The GPOCs shall run an End-to-End for every available test in the MOP TPS with the SIB.
- 10.3.4 The GPOCs shall run an End-to-End for every available test in the SBRA TPS with the SIB.

10.4 Visual Inspections

The GPOCs shall complete the same visual inspections as the Contractor on all ITAs.

10.5 ITA Acceptance & Acceptance Testing Reports

10.5.1 General Acceptance Test Report Requirements

- 10.5.1.1 The GPOCs shall generate Acceptance Test Reports for every ITA.
- 10.5.1.2 The test reports must explicitly state what tests were performed on the ITA.
- 10.5.1.3 Each test must be given a clear title and clearly labeled “PASSED” or “FAILED,” with an overall “PASSED” result only given to an ITA when no tests fail.
- 10.5.1.4 When a test fails, details of the ITA’s failure shall be written under the appropriate test header.
- 10.5.1.5 Multiple Acceptance Test Reports shall not be generated for an individual ITA. Anytime an ITA fails a test and is repaired, the new tests that are performed shall be appended to the end of the ITA’s Acceptance Test Report thus creating a continuous history of the ITA as it is tested by the GPOCs.
- 10.5.1.6 An ITA shall only be accepted by the GPOCs if and only if the overall result of its respective Acceptance Test Report is “PASSED.”
- 10.5.1.7 A digital copy of the Test Report shall be sent by e-mail to the Contractor within one business day of the ITA completing testing.

- 10.5.2 Acceptance Testing Pass Criteria – In the event of an ITA having passed every test during Acceptance Testing, the GPOCs shall sign off on the Acceptance Test Reports for the ITA. At which point the ITA becomes Government Property.

- 10.5.3 Acceptance Testing Failure Criteria – In the event of an ITA failing any tests during Acceptance Testing, the GPOCs shall then ship the ITA back to the Contractor within five business days of the ITA’s failure. Following the Contractor’s receipt of the failed ITA, they shall have twenty business days to repair it, reperform Qualification Testing, and ship it back to the GPOCs. The GPOCs shall then recomplete every test performed during Acceptance Testing in the ITA. The process detailed in this section shall be completed every time an ITA fails Acceptance Testing until it successfully completes Acceptance Testing.

11 OTHER DELIVERABLES

The following items must be delivered to the GPOCs no later than ten business days after successful acceptance testing of the final ITA produced by the Contractor—unless otherwise specified within a deliverable’s respective section.

11.1 Final Technical Data Package (CDRL 005, Appendix H)

- 11.1.1 If any changes to any of the ITAs occurred after delivery of the Preliminary Data Package, then those changes must be documented and changed accordingly on the ITA’s respective drawing.
- 11.1.2 If any new drawings related to the ITAs are generated after delivery of the Preliminary Data Package, then they must be included in the Final Technical Data Package.
- 11.1.3 Every drawing that was delivered as a part of the Preliminary Data Package must be delivered with the Final Technical Data Package—regardless of it was revised or not.
- 11.1.4 The Final Technical Data Package must be delivered through DoD Safe.

11.2 Project Residuals (CDRL 006, Appendix I)

- 11.2.1 All hardware and technical data procured for and relevant to all ITAs, their construction, maintenance, modification, and production shall all be shipped and/or e-mailed by the Contractor to the GPOCs.
- 11.2.2 All physical items delivered in this requirement must be documented in an itemized list in a Word Document (.docx file format) and sent by e-mail to the GPOCs later than one business day after shipping.
- 11.2.3 All digital items delivered in this requirement must be sent through DoD Safe to the GPOCs. An itemized list in a Word Document (.docx file format) must be sent in the same e-mail as the one in Section 11.2.2.

11.3 Unlimited Government Rights

- 11.3.1 When the Preliminary Technical Data Package and Final Technical Data Package are submitted by the Contractor, the Government shall also receive unlimited rights to use, modify, reproduce, distribute, and destroy all drawings and other digital data related to the ITAs.
- 11.3.2 When the GPOCs accept an ITA as government property they shall also receive unlimited rights to use, reproduce, modify, maintenance, and destroy the ITA.
- 11.3.3 All of the project residuals received by the GPOCs shall become government property with unlimited rights to use, reproduce, modify, and destroy the residuals.

12 SUMMARY OF REQUIRED DELIVERABLES

<u>CDRL</u>	<u>Submissions Timeline</u>	<u>Paragraph</u>	<u>Requirement</u>	<u>Responsible Party</u>
001	No more than five business days after contract award.	3	Contractor Points of Contact	Contractor
002	No more than forty business days after contract award.	6.1	Preliminary Technical Data Package	Contractor
003	No more than forty-five business days after contract award.	6.2	Design Review Presentation	Contractor
	No more than sixty business days after the GPOC approval date for the Preliminary Data Package.	7.1.1	First Article of the BRU-44 Cable	Contractor
	No more than sixty business days after the GPOC approval date for the Preliminary Data Package.	7.1.2	First Article of the PRIB	Contractor
	No more than sixty business days after the GPOC approval date for the Preliminary Data Package.	7.1.3	First Article of the SIB	Contractor
	Starting the month after Acceptance Testing of the First Article BRU-44 Cable, one BRU-44 Cable constructed every month for eight months.	7.2.1	Eight Full-Rate Production BRU-44 Cables	Contractor
	Starting the month after Acceptance Testing of the First Article PRIB, one PRIB constructed every month for nine months.	7.2.2	Nine Full-Rate Production PRIBs	Contractor
	Starting the month after Acceptance Testing of the First Article SIB, one SIB constructed every month for thirteen months.	7.2.3	Thirteen Full-Rate Production SIBs	Contractor
	Every ITA shall be Qualification Tested within ten business days of the ITA's production date.	8	Thirty-Three Qualification Tests	Contractor
004	Every ITA shall have one Qualification Test Report e-mailed to the GPOCs within one business day of successful Qualification Testing of the ITA.	8.5	Thirty-Three Qualification Test Reports	Contractor
	Dates for the test event shall be proposed by the Contractor during Design Review.	9	Temperature Testing Event	Contractor
005	No more than ten business days after the successful completion of the Acceptance Test of the final ITA.	11.1	Final Technical Data Package	Contractor
006	No more than ten business days after the successful completion of the Acceptance Test of the final ITA.	11.2	Project Residuals	Contractor
	Whenever an ITA, Technical Data Package, or project residual is accepted by the GPOCs.	11.3	Unlimited Government Rights	Contractor

Table 1. Summary of Required Deliverables

13 LIST OF ACRONYMS & INITIALISMS

APEX	(Neither an acronym nor an initialism.)
ANSI	American National Standard Institute
AWS	American Welding Society Incorporated
BRU	Bomb Rack Unit
CB	Circuit Breaker
CDRL	Contract Data Requirements List
CLIN	Contract Line Item Number
COLT	Common Organizational Level Tester
CPOC	Contractor Point of Contact
D	LED
LED	Light-Emitting Diode
GFE	Government Furnished Equipment
GPOC	Government Point of Contact
MAC	(Neither an acronym nor an initialism.)
IAW	In Accordance With
IPC	Global Electronics Association
ITA	Interface Test Adapter
MIT	Massive Ordnance Penetrator Interface Test
MIL-STD	Military Standard Specification
MIL-SPEC	Military Specification
MIL-PRF	Military Performance Specification
MMS	Mini-Mockup Simulator
MOP	Massive Ordnance Penetrator
NEMA	National Electrical Manufacturers Association
PDCU	Portable Display Control Unit
PRIB	Powered Rotary Launcher Assembly Interface Box
RLA	Rotary Launcher Assembly
SBRA	Smart Bomb Rack Assembly
SBRC	Smart Bomb Rack Controller
SIB	Smart Bomb Rack Assembly Interface Box
SOW	Statement of Work
TIA	Telecommunications Industry Association (TIA)
TITAN	(Neither an acronym nor an initialism.)
TPS	Test Program Set
UUT	Unit Under Test